```
blocked_bad = after.loc[after["is_laundering"] == 1, "amt_paid"].sum
blocked_good = after.loc[after["is_laundering"] == 0, "amt_paid"].su

print("rings used:", len(detect))
print("laundering payments blocked:", int((after["is_laundering"] ==
print("laundering money blocked:", round(blocked_bad))
print("share of all laundering money:", round(100 * blocked_bad / to
print("normal money wrongly blocked:", round(blocked_good))
```

```
rings used: 119
laundering payments blocked: 393
laundering money blocked: 2921528300
share of all laundering money: 1.56 %
normal money wrongly blocked: 25341397071
```

Varshini_ | Churn_S | clients.c | Digit_Cl | Untitled. | Fraud_L | krash.ipy | Lab01_o

Code

```
print( 'recall: ', round(recall_score(y_test, pred), 3))
```

```
AUC: 0.842
precision: 0.042
recall: 0.719
```

[6]:

```
for t in [0.5, 0.7, 0.8, 0.9, 0.95]:
    pred_t = (proba > t).astype(int)
    print("cutoff", t,
          "| flagged:", int(pred_t.sum()),
          "| precision:", round(precision_score(y_test, pred_t, zero_division=0), 3),
          "| recall:", round(recall_score(y_test, pred_t), 3))
```

```
cutoff 0.5 | flagged: 32471 | precision: 0.042 | recall: 0.719
cutoff 0.7 | flagged: 16718 | precision: 0.059 | recall: 0.514
cutoff 0.8 | flagged: 7964 | precision: 0.078 | recall: 0.325
cutoff 0.9 | flagged: 892 | precision: 0.145 | recall: 0.068
cutoff 0.95 | flagged: 66 | precision: 0.424 | recall: 0.015
```

Tabs Settings Help

Varshini_work.ipynb | Churn_Scikit_PyTorch.ipynb | clients.csv | Digit_Class_TensorFlow.ipynt | Untitled.ipynb | Fraud_LSTM_Keras_TF.ipynb | krash.ipynb | Lab01_original.ipynb

Code

```
Table: X
Rows: 515,088
Columns: out_count | fan_out | out_amt | in_count | fan_in | in_amt

Table: X_test
Rows: 154,527
Columns: out_count | fan_out | out_amt | in_count | fan_in | in_amt

Table: X_train
Rows: 360,561
Columns: out_count | fan_out | out_amt | in_count | fan_in | in_amt

Table: a
Rows: 2,807
Columns: timestamp | from_bank | from_acct | to_bank | to_acct | amt_received | recv_currency | amt_paid | pay_currency | pay_format | is_laundering | src | dst | ring | detect

Table: after
Rows: 35,613
Columns: timestamp | from_bank | from_acct | to_bank | to_acct | amt_received | recv_currency | amt_paid | pay_currency | pay_format | is_laundering | src | dst | ring | detect

Table: alerts
Rows: 7,964
Columns: account | risk | label | token | risk_flag

Table: df
Rows: 5,078,345
Columns: timestamp | from_bank | from_acct | to_bank | to_acct | amt_received | recv_currency | amt_paid | pay_currency | pay_format | is_laundering | src | dst

Table: edges
Rows: 8,501
Columns: src | dst | t_src | t_dst

Table: ex
Rows: 12
Columns: token | ring | bank | label | account | risk

Table: feat
Rows: 515,088
Columns: out_count | fan_out | out_amt | in_count | fan_in | in_amt | label | bank

Table: flow
Rows: 36
Columns: timestamp | from_bank | from_acct | to_bank | to_acct | amt_received | recv_currency | amt_paid | pay_currency | pay_format | is_laundering | src | dst | from_token | to_token

Table: inc
Rows: 420,640
Columns: in_count | fan_in | in_amt

Table: link
Rows: 433
Columns: src | dst | timestamp | ring

Table: m
Rows: 488
Columns: token | ring | bank | label | account | risk

Table: mem
Rows: 488
Columns: token | ring | bank | label | account

Table: members
Rows: 488
Columns: token | ring | bank | label | account

Table: multi
Rows: 119
Columns: accounts | banks | true_laundering

Table: nonself
Rows: 4,487,133
Columns: timestamp | from_bank | from_acct | to_bank | to_acct | amt_received | recv_currency | amt_paid | pay_currency | pay_format | is_laundering | src | dst

Table: out
Rows: 496,999
Columns: out_count | fan_out | out_amt

Table: ring
Rows: 7,596
Columns: token | ring | bank | label | account
```

```
    sel = ring_risk[ring_risk >= ring_risk.quantile(q)].index
    a = after[after["ring"].isin(sel)]
    bad = a.loc[a["is_laundering"] == 1, "amt_paid"].sum()
    good = a.loc[a["is_laundering"] == 0, "amt_paid"].sum()
    print("top", int((1 - q) * 100), "% of rings | rings:", len(sel),
          "| laundering blocked %:", round(100 * bad / total_bad, 2),
          "| laundering money:", round(bad),
          "| normal money wrongly blocked:", round(good))
```

```
top 100 % of rings | rings: 119 | laundering blocked %: 1.47 | laundering money: 2748999334 | normal money wrongly blocked: 16819192116
top 50 % of rings | rings: 60 | laundering blocked %: 1.47 | laundering money: 2746255787 | normal money wrongly blocked: 2251612018
top 25 % of rings | rings: 30 | laundering blocked %: 1.47 | laundering money: 2745991533 | normal money wrongly blocked: 475289074
```